Water Into Dry Riverbeds

A flow architecture for land, stewards, and planetary commons

Public working report · v0.4.3 · 18 October 2026
Publisher: Spiralweb Papers
Accountable knowledge steward: Lars A. Engberg · CC BY 4.0

Editorial practice: AI-assisted and human-responsible. Method and responsibility.

papers.spiralweb.earth

Executive Summary

This report documents a practice and a working architecture, not a proven doctrine.

The core argument: commitment alone cannot sustain regenerative land stewardship. Resource flows may be absent, misdirected, too fast for local carrying capacity, or tied to demands that weaken the work. These are the institutional and economic problems examined here. Ecological conditions, conflict, access to land, and many other circumstances also matter; a flow architecture addresses part of that wider situation.

What this report provides:

- **The three-stream model** — Stream A (Land and Ecology), Stream B (Steward Viability), and Stream C (Coordination and Governance), kept separately readable under a three-stream non-cancellation rule. Elir and Elia remain the relational names of the land and steward streams in the poetic layer of the work.

- **The bio-regulator** — a human-held decision-support framework for reading ecological and governance absorptive capacity before capital enters a site. It informs human judgment; it does not make or execute a funding decision.
- **A field-area calibration method** — identifying the evidence, comparison, disclosure, and local correction required before stewardship compensation or a field budget can be agreed.
- **A participatory-budgeting cycle** — a locally adaptable fiscal governance cycle that must protect real decision rights, visible reasoning, dissent, privacy, and safeguards against centralisation, patronage, and elite capture.
- **Institutional precedents and critical-friend scrutiny** — existing arrangements that slow extraction, and the assumptions most likely to fail.
- **A bridge to the current Spiralweb inquiry** — the five optional PG Ledger formats, the Penguin Dashboard, SRIP, and the public inquiry developed in Report 06.

For whom:

- Funding bodies approaching regenerative land partnerships.
- Stewardship circles establishing or formalising their governance.
- Researchers in commons governance, participatory economics, and ecological finance.
- Municipal and policy actors asking what a bioregional value architecture looks like in practice.

Key innovation: the three-stream separation helps make institutional capture visible and harder through structural non-cancellation and no silent cross-stream substitution. Land and ecology funding cannot silently change purpose. Steward support cannot purchase land authority. Coordination funding cannot accumulate invisibly as a power reservoir. The streams communicate, but conversion between them requires explicit, recorded human governance.

Preamble

Regenerative land work joins ecological observation, practical care, local knowledge, livelihood, coordination, and decisions about shared resources. The visible intervention may be planting, water retention, habitat repair, or food production. Less visible are the people who hold continuity, organise the work, maintain relationships, and keep a usable record of what changes over time.

Work of this kind is often expected to continue without a financial architecture capable of supporting the land, the people, and the coordination that holds the relation together. That is the condition this report addresses — not abstractly but precisely. The question is not whether regenerative stewardship is valuable. The question is what kind of flow architecture makes it viable without capture, hidden depletion, or flood.

> The dry riverbed is not a failure. It is a system waiting for the conditions that make flow viable without flood.

This is Report 03 in Series III. It builds on Report 01 and Report 02. Reports 04–06 have since developed the architecture further: the Penguin Dashboard supplies the monthly human governance reading; SRIP describes the bounded field-facing entry; and Regenerative Reciprocity develops the wider relational and economic inquiry. Report 03 retains its distinct contribution: the concrete flow, governance, and calibration architecture through which land, steward viability, coordination, and participatory allocation can be thought together.

Spiralweb Papers is the publisher of this independent public inquiry. Publication does not make the report policy of the Spiralweb Stewardship Association, activate a field, create a funding commitment, or appoint an institutional counterparty. If the Association or another body becomes an actual party to a field agreement, its lawful governance, accounting, and decisions apply within that bounded role.

Position of this v0.4 edition

This revision preserves the report's three-stream, flow-regulation, and participatory-governance architecture while bringing its compensation logic into line with Spiralweb's current working practice. Stewardship compensation is no longer derived from a basic-living-cost formula with a fixed activity uplift.

The report now uses field-area calibration: actual functions and time, applicable law, locally relevant professional comparators, working conditions, living-wage evidence, and correction by each person carrying the work. Living-cost evidence remains an input, not the target by itself. No method or published figure creates an offer, entitlement, rate, field activation, or payment commitment.

1. From Re-embedding to Flow

1.1 The underlying question

How do people care for shared life when inherited structures of control and certainty no longer provide a sufficient frame? Communities facing institutional erosion often do more than seek formal influence. They rebuild the feedback loops through which shared conditions can become visible, discussable, and open to correction.

Moral Biology brings the body into this inquiry. Work, insecurity, and institutional demands can affect what people have the capacity to notice, discuss, and carry through. A strained person may still judge accurately and act responsibly; strain is a reason to examine the conditions of the work, not a diagnosis of the person or a single explanation of political behaviour.

1.2 The shift to Moral Biology

Moral Biology begins from a non-negotiable premise: humans are social mammals whose ethical capacities are shaped by embodied regulation, relation, and material conditions, not by abstract rules alone. The implication for value architecture is direct. Money flowing into a governance system that cannot register the difference between nourishment and overload will not repair the system. It will amplify its pathologies.

Financial architecture can serve life only when the governance architecture can notice ecological deterioration, human depletion, conflicted authority, uncertainty, and the need to pause.

1.3 Why this report now

Three conditions make this report necessary. First, bounded field cycles can test a flow architecture, while no paper, score, or model can declare a relationship mature or active. Readiness must be established through a real place, real people, consent, recurrence, evidence, and a governance rhythm that can carry correction.

Second, participatory-budgeting and commons-governance research allow the institutional conditions and failure modes to be named more precisely. Third, regenerative-finance experiments have made the core danger clear: liquidity exceeding accountability can reorganise a living field around capital rather than ecology.

The model below is therefore a restraint architecture. It is designed to slow flow, separate unlike purposes, protect steward viability, and keep consequential judgment with named people.

2. Three Streams, Kept Separate

The architecture begins with a structural discipline: three value channels that must not disappear into one blended impression or budget. The streams share evidence and affect one another, but strength in one cannot cancel harm in another. This is the load-bearing wall.

2.1 Stream A — Land and Ecology (Elir)

Stream A supports ecological regeneration: soil, water retention, habitat, biodiversity, biomass cycles, food, tools, and locally relevant land work. In the original relational language of this report, Elir names the stream's orientation toward land and living systems.

Stream A cannot be converted silently into personal wealth, speculative claims, or narrative control. Its purpose is documented in the financial ledger and read against place-bound observation. It moves like groundwater: gradually, in relation to what a field can absorb.

The bio-regulator asks a bounded question before a proposed release: what input does the place appear to need now, what evidence supports that interpretation, what steward capacity is required, and can the current governance hold the consequences? It does not calculate permission. It organises information into a proposed flow envelope that people may accept, amend, postpone, or reject.

ARTIFACT 2.1 — Bio-Regulator Decision-Support Model (simplified)

Questions before a proposed Stream A release

- What labour, materials, and restoration inputs are locally evidenced?
- What ecological conditions matter for this place, season, and intervention?
- What remains unknown, contested, or too early to interpret?
- Can the responsible people hold the proposed flow, including consent, roles, decisions, and correction?
- Would a smaller release, a pause, or non-financial support better protect the living field?

The model is a human decision-support prompt, not a formula or composite score. No runtime computation distributes money. No algorithm or AI system makes the final judgment. A named human decision and recorded rationale are required.

2.2 Stream B — Steward Viability (Elia)

Stream B registers and supports the conditions that allow people to carry the work: livelihood viability, health, rest, protected learning and formation, continuity, and other locally relevant enabling conditions. It is not a general wage bucket. In the relational language of this report, Elia names support without capture.

Its test is not whether support merely prevents collapse, but whether the total arrangement is locally attractive, professionally recognisable, and sustainable within a lawful, bounded role. Human work carried for Stream A, B, or C is paid and recorded according to both its purpose and its legal or accounting form.

Institutions that depend on martyrdom select for those who can afford to absorb unpaid cost. They select for prior wealth, hidden support, or unsustainable self-extraction rather than competence and commitment. Stream B interrupts that selection dynamic.

2.3 Stream C — Coordination and Governance

Stream C supports the relational and institutional layer: local meetings, decisions, consent, conflict work, documentation, accounting, evidence storage, translation, partnership conditions, and the coordination required to keep the centre light and the field legible.

Where people carry this work, their compensation or service payment belongs visibly here by purpose; it is not displaced into Stream B merely because a human being receives it.

This is often the least visible stream and therefore the easiest place for power to accumulate. Stream C requires visible mandates, recorded decisions, reviewable roles, rotation where appropriate, and limits on silent institutional growth.

Format 4 holds decisions over time. Format 1 synthesises the three-stream reading. Format 5 records each financial movement and its observed effect. Public visibility may support accountability, but privacy, consent, safeguarding, and legal obligations govern what can responsibly be made public.

ARTIFACT 2.2 — Three-Stream Membrane Rules

- **STREAM A — Land and Ecology:** supports the living place and ecological work, including explicitly classified paid ecological work.
- **STREAM B — Steward Viability:** supports the conditions and protected capacity that allow people to carry the field. It cannot purchase land control or governance authority.
- **STREAM C — Coordination and Governance:** supports the holding infrastructure, including explicitly classified paid coordination and governance work. It cannot accumulate invisibly as a power reservoir.
- **Communication:** the streams share evidence and affect one another.
- **Non-cancellation:** success in one stream cannot erase deterioration in another.
- **Two-axis entry:** every payment records both its purpose stream and its transaction type, such as compensation, conditions or protection, reimbursement, procurement, or field investment.
- **No silent substitution:** any cross-stream movement requires an explicit decision, applicable consent or donor acceptance, and a recorded rationale.
- **Correction:** suspected concealment or unauthorised conversion requires proportionate human review under the actual agreement, mandate, and applicable law.

3. Gold Before Bloom

Do not open the channel before the riverbed is ready. Do not release value before the land and people can hold it. Do not scale intake faster than governance can digest. Gold Before Bloom is ecological literacy applied to institutional design.

Liquidity exceeding accountability can reorganise a field around capital rather than ecology. The architecture becomes a fund, a programme, or a reporting machine instead of a living relationship. Funds pool. Living systems differentiate and flow.

The throttles are metabolic, not ideological. Intake responds to current ecological conditions, steward viability, evidence quality, and governance capacity. When the next inflow would create overload, ordinary expansion pauses. Stabilising support, emergency care, or repair may still be appropriate; a red condition is not abandonment.

ARTIFACT 3.1 — Gold Before Bloom Checklist

Before accepting a materially larger new flow, ask:

1. What evidence supports the responsible people's current reading of governance readiness, and what remains uncertain?
2. Are the previous cycle’s ecological observations stable, improving, or honestly explained?
3. Have the people materially affected had a meaningful opportunity to participate, dissent, and withhold consent where consent is required?
4. Are existing Stream B and Stream C commitments fully covered?
5. Can the next intake be absorbed without outrunning the locally evidenced carrying capacity?
6. Do Formats 1 and 5 show that field condition and money-to-life effects remain legible?

If a material condition remains unresolved, pause ordinary expansion for human review. Record the reason, identify protective or stabilising support, and agree when and by whom the question will be revisited. The checklist informs judgment; it does not issue status.

4. A Human-Scale Circle as Living Bank

A human-scale circle can function as a living bank: not a bureaucratic wall, but a relational edge through which information, responsibility, and allocation move. The Circle of 13 is one fruitful form in the Spiralweb lineage. It is neither a universal optimum nor an entry requirement.

Ostrom’s research across irrigation systems, forest commons, and fisheries documented recurring conditions for durable self-governance: participants know the field, monitoring is close to practice, locally affected people can shape rules, conflict resolution is accessible, and governance is nested without being absorbed by a single centre.

Groups of roughly ten to fifteen may sometimes sit at a useful threshold: large enough for diversity and small enough for relational accountability. But the number does not create the conditions. A circle is credible when its people are real, roles are legible, differences can be voiced, affected people can halt movement, and responsibility can continue when one person pauses.

Relational authority is earned through time, care, knowledge, and accountable practice. It is not granted by title, longevity alone, funding access, or proximity to the centre. Intergenerational participation may strengthen the field when it is real, consent-based, and suited to the place.

PG Ledger and the five optional formats may make the circle's condition revisitable without requiring a remote authority to name what the field is. The wider 13×13 inquiry can help a circle ask what it has not yet noticed; it is not a hidden scoring matrix or a requirement that every circle contain thirteen people or every region contain thirteen circles.

PART III — PARTICIPATORY BUDGETING AS FISCAL GOVERNANCE

5. The Annual Budget Cycle

The governance structure requires a fiscal practice to match it. Participatory budgeting, pioneered in Porto Alegre and adapted across many cities and institutions, offers an important precedent. Its promise is not guaranteed by the label.

Research on citizen participation documents how bureaucratic and technocratic centralisation can reduce participation to a limited, unidirectional relation in which citizens are positioned largely as users. Formal participatory rights may coexist with clientelism, patronage, and networks of power that narrow real influence (Fung & Wright, 2003; Wampler, 2010).

This evidence does not invalidate participatory budgeting. It clarifies its conditions. Allocation becomes empowering only where affected people possess meaningful decision rights, information is understandable, dissent is protected, facilitation does not own the outcome, privacy and safeguarding are respected, and capture can be named and corrected.

ARTIFACT 5.1 — Locally Adaptable Participatory Budget Cycle

1. **Data assembly.** Bring together the relevant ecological observations, current three-stream reading, financial ledger, commitments, uncertainties, and unresolved decisions. Formats 1, 2, 4, and 5 may serve as working surfaces where they are locally useful and proportionate.
2. **Steward review.** People carrying the consequences read the material together. Locally trusted facilitation may be invited; it does not own the decision.
3. **Affected-community input.** Provide a bounded and accessible window for input. Record substance accurately. Do not publish personal, sensitive, or safeguarded material merely in the name of transparency.
4. **Allocation proposal.** State the proposed changes to Stream A, Stream B, Stream C, infrastructure, and reserve. Link each change to evidence, uncertainty, and the people expected to carry it.
5. **Decision.** Use a locally agreed rule that protects affected-person voice and guards against faction capture. No fixed threshold substitutes for consent, a material protection concern, or attention to who is absent from the decision.
6. **Publication and record.** Publish the decision, reasoning, conditions for revision, and financial classification at the level permitted by consent, safeguarding, and law. Record dissent rather than manufacturing unanimity.
7. **Rotation and review.** Facilitation, note-keeping, or budget review should rotate where the local group can carry rotation without losing necessary continuity. No office should become permanent by silence.

6. Calibrating Stewardship Compensation

Compensation can support work in any of the three purpose streams. It begins with the value of the stewardship work a bounded field cycle actually asks people to carry. It does not begin with inherited labels such as farm worker, landowner, researcher, director, or volunteer, and it does not use legal minimums or household subsistence as the target.

Each place develops a field-area compensation reference: the gross monthly full-time-equivalent value of sustained stewardship and co-inquiry within that field economy. The reference values actual stewardship time; it is not a promise of full-time work. National law and macroeconomic evidence frame the calculation. Locally relevant professional roles, working conditions, living-wage evidence, actual functions and time, and local power relations complete it. Living-wage evidence is an input, not the target by itself.

ARTIFACT 6.1 — Field-Area Calibration Sequence

1. **Name the work.** Record actual functions, time, continuity, learning, judgement, hosting, translation, documentation, risk, and accountability.
2. **Establish the lawful frame.** Verify classification, minimum protections, tax, benefits, safeguards, and the payment route.
3. **Build the local comparison.** Read living-wage evidence together with relevant professional, ecological, research, NGO, coordination, and skilled-work comparators.
4. **Make both axes visible.** Record the purpose stream and separately classify compensation, conditions and protection, reimbursements, procurement, living-field investment, and farm or livelihood income.
5. **Disclose and correct.** The responsible party states its provisional reasoning openly, then each affected steward can correct their own role, time, conditions, and ability to participate.
6. **Agree only when ready.** A rate exists only inside a lawful, funded, and bounded field agreement with review, pause, and exit conditions.

Equal standing does not require identical nominal salaries across countries or identical contracts across roles. It requires that stewardship be locally attractive and professionally recognisable, and that every difference follow named time, function, burden, continuity, or accountability rather than title, ownership, class, or credentials.

PART IV — FIELD-AREA CALIBRATION

7. From Method to a Bounded Field Budget

> This edition does not publish compensation corridors or complete field budgets. Earlier country budget tables were planning models and are superseded for operational use. A numerical hypothesis remains internal until its evidence is reviewed, the responsible party has disclosed its reasoning, affected stewards have corrected their own roles and terms, and an authorised body can fund and approve a lawful, bounded agreement.

The lawful frame, evidence set, professional comparators, conditions, and local correction must be established afresh in each place. Detailed local cells and provisional figures belong in the responsible party's internal compensation practice and in the relevant field agreement, not in this report.

7.1 Cross-Field Fairness Without False Sameness

Nominal exchange rates and purchasing-power conversions may help a responsible party plan its funding; they do not determine a person's value or local pay. Every field-area review therefore makes the same five lenses visible.

ARTIFACT 7.1 — Five Lenses for Cross-Field Review

Lens	What must be visible
Local amount	Gross local-currency value, actual time fraction, and payment rhythm.
Living-wage relationship	Distance from the best available locally relevant living-wage evidence.
Professional position	Placement among relevant local skilled, research, NGO, coordination, ecological, and specialist roles.
Conditions bundle	Leave, pension or social protection, insurance, learning time, expenses, continuity, and security.
Internal spread	Differences inside the same field and the named functional reasons for them.

A complete field budget keeps both dimensions readable: every entry has a purpose stream and a transaction type. Ecological stress, infrastructure, and contingency are costed in their own flows; they do not reduce compensation or turn a climate multiplier into a wage rule. Compensation, conditions and protection, reimbursements, procurement, living-field investment, farm or livelihood income, and coordination remain distinguishable.

PART V — INSTITUTIONAL PRECEDENTS

8. What Has Worked Before

This architecture is not invented from nothing. It converges with established traditions that have wrestled with the same tension: how to steward land and people fairly, prevent speculation, slow money, and avoid governance capture.

8.1 Steward-ownership

Steward-ownership separates voting rights from capital rights. Capital can enter a structure without acquiring control or unlimited speculative upside. The organisation belongs to its purpose rather than to investors alone. The principle directly addresses the flood problem: support need not purchase authority.

8.2 Community land trusts

Community land trusts hold land collectively, constrain resale by formula, and separate use rights from speculative ownership. Land can become a stabilised commons within a market system. Use without capture is the operative lesson.

8.3 RSF Social Finance

RSF has developed relational lending practices in which borrowers, investors, and intermediaries meet in dialogue. Money slows down. Transparency reduces speculative pressure. Returns are intended to remain modest and purpose-aligned.

8.4 Mondragon

The Mondragon federation demonstrates that cooperative forms can build internal capital systems, capped wage ratios, and reinvestment across a polycentric structure over decades. It is not without tensions, but it remains an important precedent for scale without pure extraction.

8.5 Buurtzorg

Buurtzorg organises nursing through small, self-managing teams. Compensation is stable and professional — not heroic, not speculative. The institution does not treat the sacrifice of practitioners as a hidden subsidy.

> The institution owes itself continuity. If it depends on sacrificial founders and heroic practitioners absorbing institutional cost from personal conviction, it is fragile. Structures, not virtues alone, stabilise systems over time.

PART VI — CRITICAL-FRIEND SCRUTINY

9. The Assumptions Most Likely to Fail

This architecture invites challenge. What follows is not defensive. It is the discipline of honest architecture: name vulnerabilities before they become failures, and preserve the ability to correct the model without rupture.

ARTIFACT 9.1 — Critical-Friend Checklist (annual review)

1. **Local attraction.** Does the proposed value reach the level occupied by the relevant local professional comparator set, with living-wage evidence used as an input rather than the target?
   Risk: legal compliance or a subsistence-derived uplift is mistaken for professional recognition.
   Mitigation: disclose the comparator set and let affected stewards challenge both the evidence and the proposed position.
2. **Function and conditions.** Do differences follow named time, continuity, burden, function, or accountability?
   Risk: ownership, title, class, credentials, or bargaining power silently determine value; task-linked payment becomes piece-rate optimisation.
   Mitigation: agree roles and conditions explicitly, keep unlike flows separate, and do not automate payment from indicators.
3. **Contingency adequacy.** Is the starting reserve sufficient under compound climate stress?
   Risk: sequential drought, flood, illness, or infrastructure failure exceeds the buffer.
   Mitigation: review the contingency reserve locally and agree what support other nodes can actually offer.
4. **PPP linearity.** Does purchasing-power comparison distort local cost and obligation?
   Risk: informal economies and unequal access make translation nonlinear.
   Mitigation: treat PPP as orientation, never truth or moral equivalence.
5. **Metric gaming.** Are simplified indicators displacing ecological reality?
   Risk: performance is optimised for the record rather than the place.
   Mitigation: separate observation, interpretation, judgment, and decision; use mixed evidence, uncertainty, dissent, and local expertise.
6. **Moral suspicion.** “You are paying yourselves to do good.”
   Answer: labour is real, but payment must remain legible, proportionate, conflict-aware, and governed. Virtue neither justifies payment nor makes unpaid depletion acceptable.
7. **Automation drift.** Has a model, dashboard, or AI summary begun to determine readiness or release?
   Mitigation: restore the Last Impulse: named human authority, access to evidence, affected-person voice, and a recorded rationale.
8. **Template universality.** Are observation prompts, thresholds, or a circle form being imposed where they do not fit?
   Mitigation: retain the three-stream grammar while adapting evidence, thresholds, rhythm, and participation to the place.
9. **Visibility as extraction.** Does transparency expose people, minors, sacred knowledge, conflict, or local data beyond consent?
   Mitigation: distinguish accountable records from compulsory publicity.

10. Penguin Economics: Rotation, Exposure, and the Cold Edge

Emperor penguin huddles provide the image of shared warmth used in Penguin Economics. The human proposal takes that image in a normative direction: exposure should be noticed, support should reach the exposed edge, and recovery should be possible. It is not a claim that penguins follow a rule assigning every individual an equal turn in the cold. The biological account and its limits are discussed in GP12.

Role rotation is one institutional expression of the principle: budget facilitation, oversight, and exposed leadership should move where the group can carry that movement. Sunset clauses may prevent a role from becoming permanent by default, while continuity and specialist knowledge are protected through deliberate handover.

The mature economic claim is wider. In an exposed system, capacity moves toward the cold edge. Fragile fields are protected. Stronger fields carry more when they genuinely can. Surplus circulates rather than accumulating at one node. Support follows vulnerability without rewarding dysfunction.

This is also a nervous-system principle. Burnout is not proof of commitment or a private failure. It is evidence that burden, boundary, support, or rotation has failed. Penguin Economics therefore governs money, attention, rest, visibility, risk, and institutional weight — not office rotation alone.

PART VII — CLIMATE AND THE HELICOPTER VIEW

11. Climate Stress and the Sensing Network

Each field is embedded in a climate reality that is not static. Species migrate. Habitats shift. Monsoons arrive differently. Heat, drought, flooding, and water scarcity compound. Front-line stewards experiencing these changes are not recipients of charity. They are knowledge-holders within a wider sensing network.

An observation becomes usable evidence when its place, date, source, method, uncertainty, and consent conditions remain visible. Format 2 registers field observation. Format 1 holds the recurring three-stream reading. Format 4 records decisions and revision conditions. Format 5 links money to action and observed effect. The wider 13×13 inquiry can help reveal dimensions the local record has not yet considered.

A stable budget alongside ecological improvement may invite inquiry into efficiency; it does not prove efficiency. A rising budget alongside ecological stagnation may require scrutiny; it does not by itself establish failure. Climate, time lag, baseline quality, social conditions, and intervention history matter. The record supports judgment. It does not automate causal conclusions.

> We are not governing nature. We are learning to listen more carefully, and making the conditions of that listening legible across scales.

12. 13×13 Crosswalk — Flow Architecture in the Framework

The three-stream architecture can be located within the wider 13×13 inquiry documented in Knowing From the Ground. These are this paper's adaptations of the inquiry lenses discussed in Report 02 and compared with situated questions in Knowing From the Ground. The crosswalk offers revisable inquiry lenses for asking which relations and consequences need attention. It does not divide reality into thirteen fixed levels, prescribe a complete monthly matrix, or score a place. Earlier development used the language of layers and a "Planetary Operating System"; those terms describe the history of the inquiry, not the authority of this table.

Inquiry lens	Flow-architecture connection
Lens 1 — Planet / Gaia	Planetary boundaries and climate conditions orient the inquiry; they do not issue local permission.
Lens 2 — Life / Biosphere	The Land / Life reading concerns ecological processes selected for the actual place; spending through Stream A is recorded separately.
Lens 3 — Human Body	Stream B protects steward viability, rest, health, livelihood, and carrying capacity.
Lens 8 — Community	A real local circle or other credible group holds participation and responsibility.
Lens 9 — Institutions	Participatory budgeting, agreements, formats, and correction stabilise shared decisions.
Lens 10 — Economy	The three financial streams and reserve rules organise resource flow without allowing one stream to cancel deterioration in another or silently substitute for it.
Lens 12 — AI	AI may organise and compare records; it does not determine truth, colour, readiness, or allocation.
Lens 13 — Federation and emergence	Place-bound nodes may federate through shared grammar while retaining local authority and difference.

PART VIII — WE WERE NEVER EXPELLED

13. The Deeper Claim

Long-lived agroforestry, water stewardship, seed keeping, and commons practices show that regenerative land care is not a recent invention. Across generations, people have shaped living landscapes through accumulated observation of which species, rhythms, and forms of cooperation sustain one another.

Regenerative land care continues in many traditions; it has also been damaged where enclosure, colonial dispossession, extraction, and institutional change severed people's access to land and the transmission of knowledge. The histories differ. Renewing a relationship with a place begins by learning what is still carried there, who has been displaced, and which forms of care can be supported now.

What Ostrom demonstrated empirically is that communities can govern commons successfully where the conditions are present: human-scale deliberation, local monitoring, locally shaped rules, graduated accountability, accessible conflict resolution, and nested autonomy.

The contemporary addition is not a central intelligence. It is a set of light, correction-capable memory surfaces. PG Ledger, Formats 1–5, and the 13×13 inquiry can make observations, decisions, uncertainty, burden, and money revisitable across sites that may never meet.

AI can help knowledge workers and stewards organise observations, translate language, compare records, and identify missing questions. Direct observation must remain distinguishable from AI-assisted interpretation. The person or group making a consequential claim retains the Last Impulse: authority to change or reject the output, access to the relevant evidence, and responsibility for the decision.

> One day at a time. One place at a time. One season of honest evidence at a time.

Conclusion

What this report has argued:

- The three-stream separation — Stream A, Stream B, and Stream C, with Elir and Elia as relational names — prevents one form of apparent success from hiding another form of harm.
- Gold Before Bloom is the metabolic principle. Capital intake must not exceed the absorptive capacity of land, people, evidence, and governance.
- The bio-regulator is a decision-support model. It does not enforce flow technically or replace a named human judgment.
- A human-scale circle can function as a living bank. The Circle of 13 is one possible convening form, not a universal requirement.
- Participatory budgeting can become fiscal governance only where decision rights, evidence, dissent, privacy, and counter-capture safeguards are real.
- Field-area calibration begins with evidence and local correction, not a published country table. A numerical corridor becomes usable only through a bounded agreement.
- Martyrdom is not a sustainable institutional mechanism. Steward viability is mission-critical infrastructure.
- Penguin Economics distributes exposure and capacity: support moves toward the cold edge, and genuine surplus does not stop at the first strong node.

What comes next:

The model needs to be tested through real, bounded cycles. Budget figures require local repricing. The participatory cycle must be run with people carrying the consequences. Formats 1–5 must show whether observation, decision, and money remain connected without becoming burdensome. Any future operational guidance must distinguish what can become a shared procedure from what must remain local judgment.

The first complete cycle should be documented honestly: what worked, what broke, what was too heavy, what was missing, what needed less structure, and what required more. Front-line stewards must be co-designers, not recipients of a finished architecture.

Because they are already doing the work. The architecture is the part still being tested.

References

Engberg, L. A. (2025). Green Paper 12 — Penguin Economics: Rotation as Care. Spiralweb. papers.spiralweb.earth/papers/penguin-economics

Engberg, L. A. (2026). Report 01 — Kommunalt Arbejde som Natur (v1.2.4). Spiralweb Papers. papers.spiralweb.earth/papers/kommunalt-arbejde-som-natur

Engberg, L. A. (2026). Report 02 — The Correction Loop: Preserving Human Accountability in AI-Assisted Work (v1.1). Spiralweb Papers. papers.spiralweb.earth/papers/the-correction-loop

Engberg, L. A. (2026). Report 04 — Penguin Dashboard: Legibility as Governance (Version 05). Spiralweb Papers. papers.spiralweb.earth/papers/penguin-dashboard-legibility-as-governance

Engberg, L. A. (2026). Report 05 — SRIP: The Steward's Journey (v0.2). Spiralweb Papers. papers.spiralweb.earth/papers/srip-the-stewards-journey

Engberg, L. A. (2026). Report 06 — Regenerative Reciprocity (v0.8). Spiralweb Papers. papers.spiralweb.earth/papers/report-06-regenerative-reciprocity

Spiralweb Papers. (2026). Knowing From the Ground. papers.spiralweb.earth/papers/knowing-from-the-ground

Fung, A., & Wright, E. O. (2003). Deepening Democracy: Institutional Innovations in Empowered Participatory Governance. Verso.

Ostrom, E. (1990). Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge University Press.

Ostrom, E. (2005). Understanding Institutional Diversity. Princeton University Press.

Ostrom, V. (1997). The Meaning of Democracy and the Vulnerability of Democracies. University of Michigan Press.

Purpose Foundation. (2021). The Steward-Ownership Handbook. Purpose Stiftung.

RSF Social Finance. Relational-finance and lending documentation. rsfsocialfinance.org

Spiralweb. (2026). Operational Formats. spiralweb.earth/formats

Wampler, B. (2010). Participatory Budgeting in Brazil: Contestation, Cooperation, and Accountability. Penn State University Press.

Revision note

Version 0.4 made the report a generic, independent public inquiry: named field participants and place-specific examples were removed; publisher and institutional roles were separated; and fixed simulation thresholds, automatic status effects, and formula-like readiness language were

withdrawn. The three-stream, Gold Before Bloom, field-area calibration, participatory-governance, and human-accountability architecture is preserved. Version 0.4.1 shortens the process disclosure without changing that architecture. The report creates no field status, rate, mandate, or payment commitment.

Suggested citation:

Engberg, L. A. (2026). *Water Into Dry Riverbeds: A Flow Architecture for Land, Stewards, and Planetary Commons*. Report 03, public working report (v0.4.3, 17 October 2026). Spiralweb Papers. CC BY 4.0.

papers.spiralweb.earth

Publisher: Spiralweb Papers · CC BY 4.0